

Quotient Laws of Exponents

Teacher's Edition

Read This First

This lesson rebuilds Math Nation 3.2 (SE pp. 106–111). The book runs three rotating partner stations — quotient of powers, power of a quotient, zero exponent — and then a matching activity. All three discoveries are here as Notes I–III, taught from the front with the same expansions the stations used.

The zero exponent is taught two ways on one slide on purpose: the pattern table (each column divided by the base) and the quotient $2^3/2^3$. The B1G-M for 8.NSO.1.3 names 'a base to the zero power equals zero' as the misconception to expect; both roads end at 1, and Question 9 of the round asks for that sentence in writing.

The book's own key for this lesson checks. Nothing in it needed repair; the rebuild is about format, not mathematics.

Standard and Targets

MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational-number bases.

Clarification Whole-number exponents, rational bases. The zero exponent is inside 'whole-number exponents'; negative exponents wait for 3.04.

Boundary Bases are nonzero wherever a quotient or a zero exponent appears; the notes say so and the bank never asks about 0^0 .

Learning target I can use the quotient of powers, power of a quotient and zero exponent laws to rewrite and evaluate a numerical expression.

Essential question Why must the bases be the same to apply the quotient of powers law?

Building on Lesson 3.01 — the product laws; MA.6.NSO.3.3 — evaluating powers.

Working toward MA.8.NSO.1.3 — the same laws with negative exponents (3.04–3.07), where $2^3/2^5 = 2^{-2}$ is the step after today's $2^5/2^3 = 2^2$.

Where this sits in the course Second of the three MA.7.NSO.1.1 lessons. 3.03 combines today's laws with yesterday's in one expression.

Lesson at a Glance

Windy Hill periods run **53 minutes**. The whiteboard round takes the remainder after the taught blocks — **19 minutes** in this lesson — and IXL takes the last 5. Slide numbers below are read from the deck when this edition is built.

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2-3	5 min
Notes	4-6	11 min
Example 1	7-8	3 min
Your Turn	9-10	3 min
Example 2	11-12	3 min
Your Turn	13-14	3 min
Whiteboards	15-32	19 min
IXL	33	5 min
Total		53 min

Materials Whiteboards and markers. Calculators are allowed on everything; $5^6 = 15,625$ is on the notes for the value check.

Where the lesson lives Notes I (the cancelling expansion) and Notes III (the two roads to $2^0 = 1$). Whiteboard 5 and 6 are the diagnostics: a board full of B on Q5 is the zero-equals-zero misconception, a board full of B on Q6 is dividing the exponents.

No partner or group work anywhere in this lesson. Every Math Nation exploration this lesson draws on was written for pairs; here the same tables are worked from the front and on individual boards.

Vocabulary

quotient of powers law — $\frac{a^m}{a^n} = a^{m-n}$, $a \neq 0$ — same base, subtract the exponents

power of a quotient law — $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$, $b \neq 0$ — the exponent goes to numerator and denominator

zero exponent law — $a^0 = 1$ for every nonzero a

Warm-Up — spaced retrieval

Four questions, two minutes silent, then the reveal. No reteach longer than one sentence per question.

1. Write as one power: $3^2 \cdot 3^3$ **3^5**
retrieves: yesterday — 3.01 — product of powers

2. $\sqrt[3]{-27}$ **-3**
retrieves: last week — Unit 2 — cube roots (2.03)

3. Solve $2x + 3 < 11$ **$x < 4$**
retrieves: last unit — Unit 1 — two-step inequalities (1.08)

4. Simplify $\frac{5 \cdot 5 \cdot 5}{5 \cdot 5}$ **5**

Slide-by-Slide Teaching Notes

Slide 1. Quotient Laws of Exponents 1 min

- Post the learning target. Say the 'today' line and nothing else yet.

Slide 2. Warm-Up — On your own. Four minutes. No notes. 4 min

- Q4 is the move the whole lesson runs on: cancel a factor of 5 from top and bottom, and one 5 is left. A board showing $125/25$ unsimplified has the arithmetic and not the idea.

Slide 3. Warm-Up — Answers. 1 min

- Reveal. Ask for the band each question came from only if time allows.

Slide 4. Notes — Where the rule comes from. 4 min

- Expand 5^6 over 5^3 , cancel three pairs, three 5s remain. Then the rule. Ask why the base must be nonzero (division by zero) and why the bases must match (nothing cancels in $5^6/2^3$).
- **OFF THE SLIDE, YOURS TO SAY:** 'Subtract, do not divide. 5^6 over 5^3 is 5^3 , not 5^2 .' Question 6 in the round has 4^4 waiting for the divider.

Slide 5. Notes 3 min

- Three copies of $8/10$, multiplied: $8 \cdot 8 \cdot 8$ over $10 \cdot 10 \cdot 10$. The reveal on the line is the law. The task 'apply the law, do not simplify' exists because students want to reduce $8/10$ first — both routes are right and give 0.512, and that is worth saying once.

Slide 6. Notes 4 min

- Road 1: the pattern table, each column divided by the base. Road 2: $2^3/2^3$ is 1 because anything nonzero over itself is 1, and it is 2^0 by the quotient law — so $2^0 = 1$. The misconception is $a^0 = 0$; the table kills it because the next entry after 2 is $2 \div 2$, not $2 - 2$.
- **OFF THE SLIDE, YOURS TO SAY:** 'Zero exponent means NO factors of the base are multiplied in. What is left is the 1 you always start from.'

Slide 7. Example 1 — A negative base. 1 min

- SPOKEN ONLY. Same base? Yes, (-2) . Which law? Then ask what the sign of the answer will be before anyone computes.

Slide 8. Example 1 — Worked. 2 min

- Subtract, keep the base with its sign, evaluate: three negative factors, negative answer. The check row divides 64 by -8 . A student who gets $+8$ dropped the sign at the last step.

Slide 9. Your Turn — Same steps, your numbers. Boards up when done.

2 min

- Catches a divider ($3^3 = 27$ from $6 \div 2$) and a student who wrote 1^4 by dividing the bases. 81 or 3^4 is right.

Slide 10. Your Turn — Answer. *1 min*

- Reveal; name what a wrong board most likely did (see the note above).

Slide 11. Example 2 — A decimal base inside a quotient. *1 min*

- SPOKEN ONLY. Two routes: apply the power of a quotient, or divide first ($0.2 \div 10 = 0.02$) and raise that. Ask which they would choose and why.

Slide 12. Example 2 — Worked both ways. *2 min*

- Route 1 keeps the numbers small until the last step. Route 2 is one power of a small decimal; a calculator gives 0.00000016 either way. Count the decimal places on the board: 0.0016 has four, 10^4 moves four more.

Slide 13. Your Turn — Same steps, your numbers. Boards up when done. *2 min*

- Catches a student who cubes only the numerator ($27/4$) or only the denominator ($3/64$). $27/64$ is right; 0.421875 is the same value and also right.

Slide 14. Your Turn — Answer. *1 min*

- Reveal; name what a wrong board most likely did (see the note above).

Slide 15. Whiteboards · Question 1 of 9 *19 min for the round*

- Quotient of powers, plain case. A board with 10^8 added; 1^2 divided the bases.

Slide 16. Whiteboards · Question 1 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 17. Whiteboards · Question 2 of 9

- Now evaluated. $8^4 \div 8 = 4096 \div 8 = 512$ is the check. A board with 4096 ignored the denominator; $8^4 \div 1$ is the same slip.

Slide 18. Whiteboards · Question 2 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 19. Whiteboards · Question 3 of 9

- One factor left, with its sign. A board with 6 dropped the sign; a board with 1 divided the bases.

Slide 20. Whiteboards · Question 3 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 21. Whiteboards · Question 4 of 9

- A rewrite, not a value. A board with $(-5)^7/10$ applied the exponent to the numerator only; a board with $-5^7/10^7$ has lost the parentheses (same value here, but insist on them — the sign matters in 3.04).

Slide 22. Whiteboards · Question 4 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 23. Whiteboards · Question 5 of 9

- Diagnostic. A board full of B is the classic; send them back to the pattern table on Notes III — 2, then $2 \div 2$.

Slide 24. Whiteboards · Question 5 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 25. Whiteboards · Question 6 of 9

- Diagnostic. B is the one to watch: $8 \div 2 = 4$ looks like the law. The expansion on Notes I is the answer — count what cancels.

Slide 26. Whiteboards · Question 6 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 27. Whiteboards · Question 7 of 9

- Decimal base. A board with $0.5^8 \div 0.5^5$ typed into a calculator gets 0.125 too — fine, but ask for the single power.

Slide 28. Whiteboards · Question 7 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 29. Whiteboards · Question 8 of 9

- The fraction is a distraction; the exponent decides. A board with 10/11 confused zero with the identity exponent.

Slide 30. Whiteboards · Question 8 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 31. Whiteboards · Question 9 of 9

- Written. The point is not 'both'; the point is the sentence 'so $2^0 = 1$ '. A student who writes 'Hannah, because $2^0 = 0$ ' has the misconception in writing, which is the best place to find it.

Slide 32. Whiteboards · Question 9 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 33. IXL — Last five minutes. 5 min

- IXL: whatever is not finished is tonight's practice; SmartScore 67.

The Whiteboard Round — what each board tells you

Nine questions, one at a time, each with its reveal. Boards down until the cue, all up together, scan the back row first, question the blank boards before the wrong ones. Question 9 is written work. Every wrong option on a multiple-choice question is a named error; the benchmark it comes from is cited.

#	Question	Answer	What a wrong board says
1	$\frac{10^5}{10^3}$	10^2	10^8 — added the exponents [confused with product of powers]; 1^2 — divided the bases [7.NSO.1.1 misconception: operating on the bases]
2	$\frac{8^4}{8^1}$	512	4096 — kept 8^4 and treated 8^1 as 1 [identity law misread as zero exponent]; 64 — subtracted $4 - 1$ wrong or used 8^2
3	$\frac{(-6)^6}{(-6)^5}$	-6	6 — dropped the sign of the base; 1 — divided the bases [7.NSO.1.1 misconception]
4	$\left(\frac{-5}{10}\right)^7$	$\frac{(-5)^7}{10^7}$	$(-5)^7/10$ — exponent applied to the numerator only [7.NSO.1.1 misconception: exponent applied to part of a base]; $(-0.5)^7$ — simplified first (correct value, but not what was asked)
5	Which is the value?	A — 1	(B) zero exponent read as 'nothing', so 0 [B1G-M 8.NSO.1.3 misconception: a base to the zero power equals zero]; (C) confused the zero exponent with the identity exponent, 9^1 [7.NSO.1.1 law names]; (D) treated the zero exponent as a reciprocal, which is what a NEGATIVE exponent does [misconception carried into 8.NSO.1.3]
6	Which expression is equivalent?	A — 4^6	(B) divided the exponents, $8 \div 2$ [7.NSO.1.1 misconception: 'divide' applied to the exponents]; (C) divided the bases as well as subtracting the exponents [7.NSO.1.1 misconception: operating on the bases]; (D) added the exponents — the product law applied to a quotient [confused with product of powers]
7	$\frac{0.5^8}{0.5^5}$	0.125	0.03125 — used 0.5^5 (added instead of subtracting, then misread); 0.5 — subtracted to 1 by dividing 8 by 5 badly
8	$\left(\frac{10}{11}\right)^0$	1	10/11 — identity exponent instead of zero exponent; 0 — zero-exponent misconception [B1G-M 8.NSO.1.3]

#	Question	Answer	What a wrong board says
9	Hannah cancels $2^3/2^3$ to 1. Sebastian subtracts to 2^0 . Who is right?	<i>Both — $2^0 = 1$</i>	'Hannah, because $2^0 = 0$ ' — the zero-exponent misconception [B1G-M 8.NSO.1.3]; 'Sebastian only' — did not accept that a nonzero number over itself is 1

Question Bank — variation and answers

Variation. Questions 1–3 hold the law fixed and vary the base: whole number, then a quotient that leaves one factor, then a negative base. 5–6 hold the power-of-a-quotient law and switch which part is negative. 7–8 are the zero exponent from the pattern and then from a sentence. 9–11 ask for the law by name on decimal bases. The pattern breaks at 12c, where the quotient equals 1 and the student must see that the exponents match, and at 14, where two of the six options are NOT 1 for two different reasons. The Additional sheet mirrors this and breaks with 0^5 — a zero base, not a zero exponent.

The bank and its Additional sheet are posted, never assigned; they feed the quizzes. Answers below are the same values the keys print — both come from one source.

Question Bank

- $10^{5-3} = 10^2 = 100$
- $8^{4-1} = 8^3 = 512$
- $(-6)^{6-5} = (-6)^1 = -6$
- subtract***
- $\frac{(-5)^7}{10^7}$
- $\frac{3^5}{(-4)^5}$
- 1***
- 1 — the exponent counts factors of 9; with none multiplied in, what is left is 1 (or: $9^3/9^3 = 1$ and $9^3/9^3 = 9^0$).***
- $7.6^2 = 57.76$ — ***quotient of powers***
- 1 — zero exponent***
- $\frac{0.8^4}{1.6^4} = \frac{0.4096}{6.5536} = 0.0625$ — ***power of a quotient (or $0.5^4 = 0.0625$)***
- (a) $n = 5$ (b) $n = 3$ (c) $n = 4$***
- A A***
- A, B, C, D A, B, C and D***

Question Bank — Additional

- $5^3 = 125$
- $7^1 = 7$
- $(-4)^2 = 16$
- same***

5. $\frac{(-3)^6}{(-6)^6}$
6. $\frac{7^4}{2^4}$
7. **1**
8. **1 and 1**
9. $3.2^2 = 10.24$ — *quotient of powers*
10. $\frac{1.5^3}{0.5^3} = \frac{3.375}{0.125} = 27$ — *power of a quotient (or $3^3 = 27$)*
11. **(a) $n = 9$ (b) $n = 4$ (c) $n = 6$**
12. **A A**
13. **A, B, C, D A, B, C and D**

What the audit against Math Nation found

- Math Nation 3.2 pages verified: -8 , 81 , -343 , 0.125 , 0.00000016 , the 122-block warm-up pattern ($n + n + 2 + n^2$).
- The book's warm-up (a growing block pattern) is sound but previews nothing and retrieves nothing from this course; replaced by spaced retrieval.
- Practice L2 and Homework L2 keys both check; their items are on the banks here with the notation cleaned (parentheses on negative bases).
- The book's 3.2.5 matching activity is a partner task; its three expression sets are now Notes I-III examples.

What changed from the book

- 3.2.1 pattern warm-up → four retrieval questions.
- 3.2.2-3.2.4 stations → Notes I-III.
- 3.2.5 Bring It Together and 3.2.6 Your Turn → the two worked examples and the nine-question round.
- 3.2.7 Cool Down (write an explanation) → whiteboard Question 9, written.